

PICOFLOW · REQUIREMENTS

# PicoFlow requirements

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DOCUMENT CONTROL

|             |                                       |         |                |
|-------------|---------------------------------------|---------|----------------|
| Document ID | PCF-REQ-001                           | Version | 0.3            |
| Status      | Draft                                 | Date    | 2026-09-25     |
| Author      | Amish Chadha                          | License | CERN-OHL-S-2.0 |
| Project TRL | TRL 3 Proof of concept                |         |                |
| Repository  | github.com/BoujeeEnjinia1701/picoflow |         |                |
| Source      | docs/03-requirements.md @ uncommitted |         |                |

REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                          |
|---------|------------|--------------|-----------------------------------------------------------------------------------------------------------------|
| 0.3     | 2026-09-25 | Amish Chadha | Apply Amish's 2026-09-25 decisions (PCF-DDR-001), R15 to \$450, add R17 penstock surge, status from PCF-CAL-001 |
| 0.2     | 2026-09-25 | Amish Chadha | First measurable requirements for TRL 2, with status against the concept estimates                              |
| 0.1     | 2026-09-24 | Amish Chadha | Initial scaffold                                                                                                |

## PicoFlow requirements

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These requirements were set at TRL 2 and checked by calculation at TRL 3 in PCF-CAL-001. The status column gives the TRL 3 result; "not verifiable at TRL 3" means a bench or site test is needed. Decisions of 2026-09-25 are recorded in PCF-DDR-001: the budget (R15) is now \$450, and R17 was added for the drainage-grade penstock. Targets are still not validated with users or at a site.

The **design point** used throughout is 2.0 m of gross head (forebay water surface to nozzle centerline) and 10 L/s through two nozzles, with a 20 m run of 110 mm PVC penstock.

| ID  | Requirement                       | Target                                                                                                                                                                                      | Verification (TRL 3 or later)                  | Status (PCF-CAL-001)                                                                                                                                                      |
|-----|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1  | Operating head range              | Generates from 1.0 to 3.0 m gross head with the same runner, changing only nozzle inserts                                                                                                   | Performance calculation across the range       | Met; DC bus 19.2 V at 1.0 m                                                                                                                                               |
| R2  | Flow range                        | 5 to 15 L/s through interchangeable nozzle inserts, one or two jets                                                                                                                         | Nozzle sizing calculation                      | <b>Not met at 1.0 m</b> (10.5 L/s maximum); met at 2.0 m and above                                                                                                        |
| R3  | Output at the design point        | 80 W or more DC into the battery at 2.0 m and 10 L/s                                                                                                                                        | Efficiency chain calculation; later bench test | <b>Not met</b> , 77.2 W                                                                                                                                                   |
| R4  | Output at the lowest head         | 30 W or more DC into the battery at 1.0 m and 7 L/s                                                                                                                                         | Efficiency chain calculation                   | <b>Not met</b> , 23.9 W                                                                                                                                                   |
| R5  | Water-to-wire efficiency          | 40 % or more at the design point, from gross hydraulic power to DC into the battery                                                                                                         | Calculation; later test                        | <b>Not met</b> , 39.5 %                                                                                                                                                   |
| R6  | Daily energy                      | 1.8 kWh or more per day at the design point, running 24 h                                                                                                                                   | Calculation                                    | Met, 1.85 kWh (thin margin)                                                                                                                                               |
| R7  | Battery charging and load control | Charges a 12 V nominal battery; holds the runner within 10 % of its best speed while charging; diverts 100 % of output to the dump load within 1 s when the battery is full or disconnected | Controller design review; later bench test     | Not verifiable at TRL 3; dump load 1.9 times the largest output                                                                                                           |
| R8  | Runaway and touch voltage         | Every part survives indefinite runaway at 3.0 m head with no damage; no exposed conductor above 60 V DC in any state, including runaway                                                     | Speed and voltage calculation; design review   | Met by calculation: hardware clamp keeps the bus at 48 V or less; 76.4 V only on a double fault, inside the enclosed 100 V DC side; generator overspeed rating to confirm |
| R9  | Printable parts                   | Runner and nozzles print on a 220 x 220 x 250 mm build volume in PETG or nylon; runner prints in 24 h or less                                                                               | Model check; slicer estimate                   | Met (estimate); 200 x 200 x 50 mm, about 21 h                                                                                                                             |
| R10 | Debris tolerance                  | Intake screen aperture 6 mm or less; smallest nozzle 20 mm or more; screen self-cleans by overflow                                                                                          | Design review                                  | Met by design                                                                                                                                                             |
| R11 | Serviceability                    | Runner or nozzle inserts replaced with hand tools in 30 min or less; bearings in 60 min or less; bearings and generator above the spray zone                                                | Design review; later timed trial               | Not verifiable at TRL 3; bearings above the spray by layout                                                                                                               |
| R12 | Service life                      | Bearing L10 life 20,000 h or more; runner 1 year (8,760 h) or more before replacement                                                                                                       | Bearing life calculation; material review      | <b>At risk</b> ; bearings about $1.1 \times 10^7$ h, printed runner life unknown                                                                                          |
| R13 | Setting height                    | Nozzle centerline 0.3 m or less above normal tailwater, so little head is lost below the jet                                                                                                | Model check                                    | Met, 280 mm in the model                                                                                                                                                  |

| ID  | Requirement       | Target                                                                                                                           | Verification (TRL 3 or later) | Status (PCF-CAL-001)                                                       |
|-----|-------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------------|
| R14 | Portability       | Heaviest single item 15 kg or less; turbine unit (items 1 to 6 and 11) 25 kg or less, carried to site by two people              | Mass estimate                 | Met, 22.2 kg; generator 7.0 kg (definition to confirm, see PCF-DDR-001 N4) |
| R15 | Cost              | Turbine kit (all items except penstock and battery) \$450 or less in parts; salvaged-motor variant documented                    | Priced BOM ( bom/bom.csv )    | <b>At risk</b> , \$448.00 (\$2 margin); \$378.00 with a salvaged motor     |
| R16 | Stream protection | Takes no more than half of the dry-season stream flow and returns all water to the same stream; intake passes fish-safe overflow | Site survey; design review    | Not verifiable at TRL 3 (site-dependent)                                   |
| R17 | Penstock surge    | Peak pressure at the valve (static plus surge) 50 kPa or less at 3.0 m head, with a valve that cannot close in under 10 s        | Surge calculation             | Met, 34.7 kPa with a multi-turn gate valve                                 |

## ■ Requirements not met or at risk

- **R2 (flow range) not met at 1.0 m.** 15 L/s at 1.0 m needs two 71 mm jets on a 150 mm pitch circle, far outside Turgo practice. The practical ceiling at 1.0 m is about 10.5 L/s. Proposed, awaiting Amish: state the range as 5 to 10 L/s at 1.0 m.
- **R3, R5 (design-point output and efficiency) not met.** Pipe and branch losses are about 24 % of head, not the 10 % assumed at TRL 2. A 125 mm penstock on the 20 m run gives 84.3 W and 43.0 %. Proposed, awaiting Amish: change the design-point penstock to 125 mm.
- **R4 (1.0 m output) not met.** 23.9 W; no penstock size reaches 30 W with a 75 % runner. Proposed, awaiting Amish: relax to 20 W.
- **R12 (runner life) and R15 (cost margin) at risk.**

## ■ Budget scope

The \$450 budget covers the turbine kit: runner, shaft, bearings, coupling, generator, housing, manifold and nozzles, valve, forebay and screen, frame, equipment post, rectifier, controller, dump-load and clamp resistors, and wiring. The penstock and the household battery are site- and household-specific and are costed separately (decided by Amish, 2026-09-25, PCF-DDR-001 D1).

## ■ Assumptions

The full list is in PCF-CAL-001 section 2. The main ones are:

- Penstock: 110 mm drainage PVC (103.6 mm bore), 20 m, with a rounded entrance, screen, two bends and a gate valve; branches 90 mm.
- Nozzle velocity coefficient 0.97.
- Runner jet-to-shaft efficiency 75 % for a first printed design, below the 87 to 91 % reached in laboratory work on an optimized low-head Turgo ([Williamson et al., 2013](#)).

- Generator about 10 rpm per volt of rectified DC, 12 W fixed loss at 340 rpm, 0.8  $\Omega$ ; rectifier 1.6 V drop; buck converter 94 %.
- Drainage pipe joints good for 50 kPa, to be confirmed with the supplier.

**Safety:** R8 and R17 are safety requirements. They are met on paper only; neither the clamp nor the pipe rating has been tested.